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SUPERSEDING
Army 96-111-68A
20 June 1949

MILITARY SPECIFICATION

AURAMINE HYDROCHLORIDE

(FOR COLORED SMOKE MIXTURES)

This *specification was approved by the Departments of the Army, the Navy, and the Air Force for the w of procurement services* of *the respective Departments*

L SCOPE

1.1 This specification covers one' type of dye for use in the manufacture of colored smoke mixtures.

2. APPLICABLE SPECIFICATIONS STANDARDS DRAWINGS, AND PUBLICATIONS

2.1 The following specifications and standard, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

RR-S-366-Sieves: Standard, Testing

VV-L-791-Lubricants, Liquid-Fuels, and Related Products; Methods of Sampling and Testing.

U. S. ARMY

96-111-92-Grenade, Smoke, Colored, M18.

NAVY DEPARTMENT

General Spefications for Inspection of Material.

STANDARDS

MILITARY

MIL-STD-129-Marking of Shipments.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. — The following publication, of the issue in effect on date of invitation for bids, unless otherwise stated, forms a part of this specification :

INTERSTATE COMMERCE COMMISSION

Regulations for Transportation of Explosives and Other Dangerous Articles, etc.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

8.1 Purity-The purity of the material shall be not less than 86.0 percent when tested by both chemical and spectrophotometric analysis specified in 4.4.2.

9.2 Moisture. - The moisture content of the material shall be not more than 3.5 percent when tested as specified in 4,4.3.

3.3 Part icle size.-The particle size of the material shall be such that not less than 99 percent will pass through a No. 60, U. S. standard sleve, not less than 90 percent will pass through a No. 100, U.S. standard sieve and not less than 40 percent will pass through

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a No. 200 U. S. standard sieve when tested as specified in 4.4.4.

3.4 Apparent dmdty.-Thc apparent denalty shall be 0.35 ± 0.15 when tested as specified in 4.4,5.

3.5 Performance.—The material shall burn without flaming and shall emit a continuous cloud of yellow smoke for a period of 70 ± 20 acconds when tested as specified in 4.4.6 The smoke shall be distinguishable as to color, against a contrasting color background, at a distance of 10,000 feet on a clear day (see 6.4).

3.6 Workmanship.-All work shall be done in a neat and workmanlike manner.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Lot-A lot shall consist of material produced by one manufacturer with no change in process or materials in no more than 24 consecutive hours, providing the operation is continuous, in the event the process is a batch operation, each batch shall constitute a lot (see 6.6).

4.2 Sampling-From each of five containers taken at random from each lot, two representative specimens shall be removed as follows :

Specimen A: For physical and chemical tests-l pound (see 4.4.2 through 4.406),

Specimen B: For performance test -2 pounds (see 4,4.6).

Each specimen taken shall be placed in a clean, dry container, sealed and labeled in such a manner as to identify the specimen with the prime container and the lot represented (see 6.2 and 6.8).

4.3 Inspection.

4.3.1 *Packing and marking*- The inspector shall inspect the packing and marking of the material for compliance with section 5.

4.3.2 *Inspection procedures.-For* Naval purchases, general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

4.4 Tests,

4.4.1 *Reagents* .—Analytical reagent grade chemicals and distilled water shall be used throughout the tests, Blank determinations shall be run in parallel with each test, using the same quantities of reagents used in the test, and correcting shall be applied when significant.

4.4.2 *Purity*.

4.4.2.1 *Chemical analysis* .-Weigh, to the nearest milligram (reg.), approximately 2 grams (gm) of the specimen, wash into a 600 milliliter (ml.), beaker, using a small quantity of water, and break up all lumps with a glass rod. Add 8 ml. of concentrated hydrochloric acid and sufficient boiling water to dissolve the specimen. Filter the solution through a Gooch crucible and wash with boiling writer until the filtrate cornea through colorless, Add sufficient water to bring the volume to 300 ml., bring the solution to a boil, and continue boiling for 6 minutes. Cool slightly and add very slowly, with constant stirring, 20 ml. of 35-percent sodium hydroxide solution. Bring to a boil and boil 20 minutes, stirring constantly, Cool to below 60°. and within a period of 2 hours after cooling, filter through a tared Gooch crucible, Dry to constant weight at 100 to 150°C., cool in a desiccator, and weigh.

$$\text{Percent purity} = \frac{\text{Weight of residue} \times 113.6}{\text{weight of specimen}}$$

4.4.2.2 *Spectrophotometric analysis*. - Weigh 0.173 gm. of the specimen and transfer quantitatively to a 500-ml. beaker. Add approximately 50 ml. of ethanol to wet the specimen. Break all lumps with a stirring rod. Dilute with approximately 400 ml. of water (not warmer than 80°C stir until

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dissolved, and transfer quantitatively to a 1,000-ml. volumetric flask, dilute to the mark with water, and mix thoroughly. Transfer a 10-ml. aliquot of the solution to a 250-ml. volumetric flask, dilute to the mark with writer, and mix thoroughly. Within 1 hour after preparation, transfer a portion of the solution to a spectrophotometer cell and fill. the reference cell with water. Read the transmittance (T) of the solution at a wave length of 430 millimicrons.

$$\text{Percent purity} = -100 \log_{10} T$$

Note: The above procedure and formula apply when: (1) the spectrophotometer cell depth is 10 millimeters (mm.), (2) the transmittance (T) is expressed as a fraction less than 1, and (3) the spectrophotometer is a Hwdy-General Electric recording spectrophotometer having an effective band width of 10 millimicrons. When another instrument is used, it shall first be calibrated against a Hardy-General Electric instrument or against U. S. Bureau of Standards glare filters. When a spectrophotometer cell of different depth is used, calculate the weight of the specimen in inverse proportion.

4.4.3 *Moisture* .—Determine the moisture content of the material in general accordance with method No. 300.1.5 *of* Specification VV-L-791. Use a 50-gm. specimen *of* the material, 600 ml. of toluene as the solvent, a 1,000-ml. flask, and an oil bath *or* electric heating mantle as a source of heat.

4.4.4 *Particle Size*.- Numbers 60, 100, and 200 U. S. standard sieves, conforming to Specification RR-S-366, shall be selected for the test. Nest the sieves in order of decreasing mesh size, placing the largest mesh on top and a pan at the bottom, Weigh, to the nearest centigram (cg.), approximately 10 gm. of the material and transfer to the No. 60 sieve. Brush the material with a camel's-hair brush until no more material passes through the sieve. Remove the sieve and weigh the material left on the sieve Repeat the procedure with the material on the No. 100 and No. 200 sieves. Calculate the percentage of material passing through each sieve by subtracting from 100. percent the sum of the percentages of the material retained on that sieve and all of the larger mesh sieves used.

4.4.5 *Apparent density.*

4.4.5.1 *Apparatus*.- The apparatus shall consist of a cylinder and a glass funnel. The cylinder shall have an internal length of 11 ± 1 centimeters (cm.) and a volume of 100 ml. and may be made by cutting off a 250 ml. graduate at the 100-ml. graduation, The glass funnel shall be approximately 9 cm. in diameter at the top and shall have a stem approximately 2.5 cm. in length and 1.5 cm. in inside diameter.

4.4.5.2 *Procedure.* — Weigh the cylinder. Support the funnel so that the end of the stem is centrally located 1 cm. above the top of the cylinder, Dry, to a constant weight in an oven maintained at 70 to 75°C., a portion of the specimen sufficient to fill the cylinder to overflowing. Shake the specimen. from a paper into the funnel at a rate of approximately 60 ml. per minute. Avoid jarring the cylinder during the filling operation. If clogging occurs in the funnel, return the contents to the paper and repeat the filling until the material overflows the cylinder. Strike off the material level with the top of the cylinder and weigh the cylinder and the contents. Calculate the apparent density of the specimen by dividing the increase in weight of the cylinder by 100.

4.4.6 *Performance.* —Prepare and fire six yellow smoke grenades in accordance with Specification 96-111-92. Upon firing, the grenades thus prepared shall conform to 3.5 (see 6.4 and 6.6).

4.5 Rejection and resubmission. -If any specimen, selected as specified in 4.2, fails to conform to this specification, the lot shall be rejected. The contractor, at no expense to the Government shall have the option of having a partial or complete Analysis made

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of each container in the lot and removing the nonconforming material. The contractor may then resubmit the remaining portion of the lot for acceptance testing.

5. PREPARATION FOR DELIVERY

5.1 Packing.—The material shall be packed in steel drums conforming to Specification 37D of Regulations for Transportation of Explosives and Other Dangerous Articles, etc. Each drum shall be fitted with polyethylene bag constructed of polyethylene sheet having a nominal thickness of 0.004 inch. After filling, the bag shall be closed by knotting, tying, or heat sealing.

5.2 Marking. — Each container shall be conspicuously labeled: CAUTION — AVOID SKIN CONTACT. USE WITH ADEQUATE VENTILATION. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. — Procurement documents should specify the title, number, and date of this specification.

6.2 Inspection .Inspection will not necessarily include all specification, requirements: however, the omission of inspection does not constitute a waiver of these requirements.

6.3 Sampling and testing.-When the contractor consistently produces high-quality material and operates under a system of quality control acceptable to the Government, the Government, at its discretion, may modify, in whole or in part, the sampling and testing procedures specified herein. However, the Government reserves the right to return at any time, without previous notice to the constructor, to the sampling and testing procedures specified in this specification.

6.4 Alternate requirements. — Alternate performance requirements and tests, simulating the end use of the material, may be substituted for those specified in 3.6 and 4.4.6. However, in case of dispute, the requirements and test specified in 3.5 and 4.4.6 will govern.

6.5 Batch. — A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the find product substantially uniform.

6.6 Evaluation samples.-Because of the special nature of the equipment required in preparing colored smoke grenades, the contractor may submit to the procuring agency, samples of the material he proposes to furnish for evaluation as to performance. However, the results of such tests will be considered informative only and in no sense will they be constured as acceptance tests.

6.7 Nomenclature.—The American Chemical Society nomenclature for this dye is 4,4'-bis (dimethylamine) benzhydrylidenimine hydrochloride. The colour index number (Society of Dyers and Colourists-editor F. M. Rowe) is *655*.

Notice.-When Government drawings, specifications, or other data are used *for* any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Govern. ment may have formulated, furnished, or in any way supplied the said drawings, specifications, *or* other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporat.ion, or conveying my rights or permission to manufacture, use or sell any patented invention that may *in* any way be related thereto.

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